

Real-Time Social Media Data Pipeline with Spark Streaming & Kafka

Project Description

In this project, students will design and implement a real-time data processing pipeline using Apache Spark Streaming and Apache Kafka to handle social media data.

The objective is to simulate a real-world scenario where large volumes of unstructured data from platforms like Facebook or Twitter (X) are continuously collected, processed, and redistributed for real-time consumption.

Project Workflow

1. Data Ingestion & Filtering:
 - Use an appropriate API (e.g., Graph API)
 - Apply custom filtering logic to extract relevant posts (for example: filter by keywords or hashtags).
2. Data Publishing:
 - Publish the filtered posts into a Kafka topic.
3. Data Processing:
 - Connect Spark Streaming to consume the filtered data from the Kafka topic.
 - Apply further transformation, enrichment, or aggregation in Spark (such as wordcount).
 - Display the final posts in the console in real-time, formatted for easy reading.